JOINS OF LAWVERE–TIERNEY TOPOLOGIES IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: THE FINITE-ITERATION CRITERION

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Abstract. In a Grothendieck topos $\mathcal{E}$, every Lawvere–Tierney topology j (equivalently, every subtopos) defines the class W_j of maps inverted by its sheafification, an element of the complete lattice $\mathrm{Same}(\mathcal{E})$ of classes of morphisms closed under composition and two-out-of-three. The map $j \mapsto W_j$ preserves all meets; whether it preserves the join of two topologies was known to depend on the topos. We describe the join $W_j \vee W_{j'}$ for arbitrary j, j': it is the class of maps whose image is reached from the codomain, and whose kernel pair is reached from the diagonal, by a bounded number of alternating applications of the two closure operators, and every such map is a composite of maps of $W_j \cup W_{j'}$, so the join is the closure of the union under composition alone. Consequently $W_j \vee W_{j'} = W_{j \vee j'}$ if and only if $(jj')^k = j \vee j'$ for some k, the finite-iteration axiom that governs joins of nuclei on a Heyting algebra and on the algebra of truth values of fuzzy sets, now for the subobject classifier. For presheaf toposes and topologies of ideal type, J_D and J_E for idempotent ideals D, E, the condition is the stabilisation of the chain $DE \supseteq DEDE \supseteq \cdots$, which contains the known positive answer for finite categories and the negative answer on the natural numbers; for ideals generated by two sets of objects A, B it decides when the join of "bijective at A" and "bijective at B" is "bijective at $A \cap B$"; on the natural numbers, the join of the classes of maps bijective at the even and at the odd points is the class of maps of towers that are bijective up to a uniform shift, a proper subclass of the isomorphisms of pro-sets. For the topos of sheaves on a locale and two nuclei n, n' the condition is $(nn')^k = n \vee n'$ in the frame of nuclei; on the chain $[0, 1]$, the frame of opens of a spatial non-Alexandrov space, two nuclei with fixed sets $\{0, 1\} \cup \{1/2k\}$ and $\{0, 1\} \cup \{1/(2k + 1)\}$ have join with fixed set $\{0, 1\}$, the double-negation nucleus, but violate the condition, so the join of the two classes is strictly smaller than the class of maps inverted by the double-negation sheafification; on the real line, the classes of maps of sheaves that are isomorphisms over the rationals and over the irrationals have a join strictly smaller than the class of the meet of the two subtoposes, the obstruction being the unbounded length of an alternating Cantor–Bendixson derivation.

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1. Introduction

Let $\mathcal{E}$ be a Grothendieck topos. A Lawvere–Tierney topology j on $\mathcal{E}$, equivalently a subtopos $\mathrm{Sh}_j(\mathcal{E})$, has a sheafification functor a_j, and the class W_j of maps inverted by a_j, the j-local isomorphisms, is a *sameness*: a class of morphisms containing the identities and closed under composition and two-out-of-three. Samenesses form a complete lattice $\mathrm{Same}(\mathcal{E})$ under inclusion, meets being intersections and the join of two classes the least sameness containing both [8]. The assignment $j \mapsto W_j$ preserves all meets [10, Thm. 3.1], and the question studied in [10, 11] is whether it preserves binary joins: whether every map inverted by the sheafification of $j \vee j'$ is obtained from j-local and j'-local isomorphisms by finitely many compositions and cancellations. For presheaf toposes the answer was known to be yes on every finite category [11] and no on the poset $\mathbb{N}$ [10, Thm. 7.1]. This paper describes the join $W_j \vee W_{j'}$ for arbitrary topologies on an arbitrary Grothendieck topos and gives the exact criterion for it to equal $W_{j\vee j'}$.

A topology j has a closure operator c on subobjects, and a map $f\colon X \to Y$ is in W_j iff it is *j-bidense*: the image of f is c-dense in Y and the kernel pair of f lies in the closure of the diagonal of X. For two topologies j, j' with closure operators c, c' let U_k be the class of maps $f\colon X \to Y$ with

$$(cc')^k(\mathrm{im} f) = Y \qquad \text{and} \qquad \ker f \subseteq (cc')^k(\Delta_X),$$

the maps that are bidense *uniformly* with respect to the k-fold alternation of the two closures.

Theorem 1.1. *For any Lawvere–Tierney topologies j, j' on a topos $\mathcal{E}$,*

$$W_j \vee W_{j'} = \bigcup_k U_k.$$

U_0 is the class of isomorphisms, and for $k \geq 1$ every map in U_k is a composite of $2k$ epimorphisms followed by $2k$ monomorphisms lying alternately in $W_{j'}$ and W_j. In particular the join is the closure of $W_j \cup W_{j'}$ under composition.

Theorem 1.2. *Let $\mathcal{E}$ be a Grothendieck topos. Then $W_j \vee W_{j'} = W_{j\vee j'}$ if and only if $(jj')^k = j \vee j'$ as endomorphisms of the subobject classifier Ω for some k; equivalently, $(cc')^k$ is the closure operator of $j \vee j'$ on every subobject lattice.*

The closure operator of $j \vee j'$ is the transfinite iterate of cc' (Lemma 2.2), so Theorem 1.2 says that the join of the two classes is the class of the join topology exactly when the transfinite iteration stops at a finite stage uniformly. This is the condition $(j_2 j_1)^N = j_1 \vee j_2$ that governs joins of nuclei on a Heyting algebra and their joins in the sameness lattice of fuzzy sets [12, 14], now for the internal frame Ω. Theorem 1.1 holds in every elementary topos; Theorem 1.2 uses countable coproducts and transfinite unions of subobjects.

For a presheaf topos $\mathrm{Presh}(S)$ and Grothendieck topologies J, J' on the small category S the conditions become conditions on sieves (Lemma 5.1): the image sieves $\{b : y \cdot b \in \mathrm{im} f\}$ and the equaliser sieves $\{b : x \cdot b = x' \cdot b\}$, $f(x) = f(x')$, must be sent to the maximal sieve by $(jj')^k$. For topologies of *ideal type* the criterion becomes a chain condition. A two-sided ideal D of S is a set of morphisms closed under composition with arbitrary morphisms on either side; it is *idempotent* if every member is a composite of two members. Every idempotent ideal D defines a topology J_D whose covering sieves on c are the sieves containing $D(-, c) = \{a \colon x \to c \text{ in } D\}$ [9, Thm. 5.1] (for monoids, [4, Thm. 2.9]: the weak ideal topology of a two-sided ideal I is a topology iff $I^2 = I$); on a finite category every topology is of this form (Kelly–Lawvere [5], [7]), on an infinite one not necessarily. For ideals D, E write $DE = \{d \circ e : d \in D, e \in E\}$, again an ideal, and $(DE)^k = DE \cdots DE$ (k factors of each).

Corollary 1.3. *Let D, E be idempotent ideals of S. Then $W_{J_D \vee J_E} = W_{J_D} \vee W_{J_E}$ in* $\mathrm{Same}(\mathrm{Presh}(S))$ *if and only if the chain $DE \supseteq (DE)^2 \supseteq \cdots$ stabilises, i.e. $(DE)^m = (DE)^{m+1}$ for some m. In that case $I' := (DE)^m$ is an idempotent ideal with $(ED)^{m+1} = I'$, $J_D \vee J_E = J_{I'}$, $W_{J_D} \vee W_{J_E} = W_{J_{I'}}$, and every $J_{I'}$-local isomorphism is a composite of $4m$ maps of $W_{J_D} \cup W_{J_E}$. In every case $W_{J_D} \vee W_{J_E}$ is the class of maps locally bijective along some power $(DE)^k$.*

On a finite category the chain stabilises for every pair, which recovers [11] with a proof of a few lines in place of the separated quotients and plus towers used there; on $\mathbb{N}$ with the ideals of the even and of the odd points it is strictly decreasing, which recovers [10, Thm. 7.1]. The criterion becomes concrete for ideals generated by objects. For a set A of objects, the ideal D_A of morphisms factoring through A is idempotent, and $W_A = W_{J_{D_A}}$ is the class of maps bijective at every object of A, the class inverted by restriction to A, the sheafification for the essential subtopos $\mathrm{Presh}(A) \hookrightarrow \mathrm{Presh}(S)$. Proposition 6.1 decides when $W_A \vee W_B = W_{A \cap B}$: iff every morphism factoring through a long enough alternating zigzag of objects of B and A factors through an object of $A \cap B$, on a locally finite poset iff the strictly alternating chains avoiding $A \cap B$ have bounded size. On $\mathbb{N}$ with the even and the odd points the join is the class of maps of towers that are bijective up to a uniform shift, a proper subclass of the isomorphisms of pro-sets (Example 6.3). For a localic topos $\mathrm{Sh}(L)$ the topologies are the nuclei on the frame L, and Corollary 7.2 reads: $W_n \vee W_{n'} = W_{n \vee n'}$ iff $(nn')^k = n \vee n'$ in the frame of nuclei. Open and closed nuclei always satisfy this with $k = 1$; nuclei on the chain $[0, 1]$, the frame of opens of a spatial space whose sheaf topos is not a presheaf topos, need not (Example 7.4).

The proofs are elementary and use only the calculus of closure operators on subobjects. The inclusion $U_k \subseteq W_j \vee W_{j'}$ is a peeling argument: an epimorphism whose kernel pair lies in the k-fold alternating closure of the diagonal is divided by the equivalence relation "kernel pair intersected with

the c'-closure of the diagonal", which is a j'-local isomorphism, and the induced map needs one closure fewer; dually a monomorphism is enlarged to the c'-closure of its image. The failure direction uses, for a dense subobject S of Y that is not reached in k steps, the fold map of two copies of Y glued along S; the coproduct of these maps over all k is inverted by the join sheafification but lies in no U_k.

What is known. Samenesses and W_j are from [8]; that meets of topologies go to meets of samenesses, the reduction lemma and the $\mathbb{N}$ counterexample are from [10]; the finite case, with the alternating separated quotients, the plus tower and the retraction of representing bimodules, is [11]. Lawvere–Tierney topologies, universal closure operators and the bidense description of the maps inverted by sheafification are standard [6, V.1, V.3], [1, A4.3–A4.4]; for presheaves the description by image and equaliser sieves is [6, III.7]. Ideal topologies, the correspondence with idempotent ideals on finite categories and the local operators $R \mapsto \{a : a \cdot D \subseteq R\}$ are in [5, 7, 9]. Joins of nuclei as transfinite iterates are in [2]; for Lawvere–Tierney topologies, Khanjanzadeh and Madanshekaf [4] observe that the composite jj' is in general only a *weak* topology (not idempotent), show that the weak topologies on a cocomplete topos form a complete residuated lattice, and compute the join of two topologies as the topological reflection of their pointwise join, a transfinite iterate; Lemma 2.2 is this computation for the composite, at the level of closure operators. The maps inverted by sheafification, and joins of such classes, are not considered there. The first version of this paper (archived at DOI 10.5281/zenodo.23467281) proved Corollary 1.3 for ideal topologies only, with the stabilising direction by the route of [11]; the second version (DOI 10.5281/zenodo.23468327) proved Theorems 1.1 and 1.2 for presheaf toposes by sieve arguments.

What is new. Theorem 1.1, the description of the join for arbitrary topologies on arbitrary toposes and the fact that no cancellations are needed; Theorem 1.2, the finite-iteration criterion, which decides the question for every pair of subtoposes of every Grothendieck topos, whereas [11] proves the positive answer under a finiteness hypothesis; Corollary 1.3 with its short proof; Proposition 6.1 and Example 6.3; Corollary 7.2 and Example 7.4. The results do not follow from the theory of localisations or of the subtopos lattice alone: the meet of the two subtoposes, i.e. the join topology, is determined by j and j', but the join of the two classes in $\mathrm{Same}(\mathcal{E})$ is a finer invariant (the uniform version of bidensity), and Example 6.3 exhibits two subtoposes whose meet is the degenerate subtopos while the join of their classes is a proper sameness. In the localic case (Section 7) the meet of two sublocales is classical, for instance the meet of the sublocales $\mathbb{Q}$ and $\mathbb{I}$ of $\mathbb{R}$ is a nonempty dense sublocale [3, II.2]; what is new is the statement at the level of maps of sheaves, that the join of "isomorphism over $\mathbb{Q}$" and "isomorphism over $\mathbb{I}$" is strictly smaller than the class of the meet, and the identification of the

obstruction with the unbounded length of the alternating Cantor–Bendixson derivation $F \mapsto \mathrm{cl}(\mathrm{cl}(F \cap \mathbb{I}) \cap \mathbb{Q})$ (Example 7.6).

2. Preliminaries

Closure operators. Let $\mathcal{E}$ be a topos. A Lawvere–Tierney topology $j\colon \Omega \to \Omega$ corresponds to a *universal closure operator*: for each object X a map $A \mapsto c(A)$ on $\mathrm{Sub}(X)$ that is inflationary, monotone, idempotent, preserves finite intersections, and is stable under pullback, $g^{-1}(c(B)) = c(g^{-1}(B))$ for $g\colon X \to Y$ and $B \subseteq Y$; the closure of A is classified by $j \circ \chi_A$ [6, V.1]. Topologies are ordered pointwise, $j \leq j'$ iff $c(A) \subseteq c'(A)$ for all A, and the correspondence with closure operators is an order isomorphism. A subobject $A \subseteq X$ is *c-dense* if $c(A) = X$. We write $\mathrm{im}f$ for the image of $f\colon X \to Y$, $\ker f \subseteq X \times X$ for its kernel pair, $\Delta_X \subseteq X \times X$ for the diagonal, and $g(A)$ for the image of a subobject $A \subseteq X$ under g.

Lemma 2.1. *Let c be a universal closure operator, $g\colon X \to Y$, $A, K \subseteq X$, $B \subseteq Y$.*

(i) $g(c(A)) \subseteq c(g(A))$.
(ii) $c(A) \cap K \subseteq c(A \cap K)$.
(iii) $c(A \times B) = c(A) \times c(B)$ *in* $\mathrm{Sub}(X \times Y)$.
(iv) $c(\Delta_X)$ *is an equivalence relation on* X.

The same holds for every composite w of universal closure operators, which is again inflationary, monotone, pullback-stable and intersection-preserving.

Proof. (i) $A \subseteq g^{-1}(g(A))$, so $c(A) \subseteq c(g^{-1}(g(A))) = g^{-1}(c(g(A)))$. (ii) Let $m\colon K \hookrightarrow X$; then $A \cap K = m(m^{-1}(A))$ and $c(A) \cap K = m(m^{-1}(c(A))) = m(c_K(m^{-1}(A))) \subseteq c(m(m^{-1}(A)))$ by (i), where c_K is the closure on $\mathrm{Sub}(K)$, the pullback of c. (iii) $A \times B = \pi_1^{-1}(A) \cap \pi_2^{-1}(B)$, and c preserves intersections and pullbacks. (iv) Reflexivity and symmetry are clear (the twist map preserves Δ_X and commutes with c). The composite of $c(\Delta_X)$ with itself is $\pi_{13}\big(\pi_{12}^{-1}c(\Delta_X) \cap \pi_{23}^{-1}c(\Delta_X)\big) = \pi_{13}\big(c(\pi_{12}^{-1}\Delta_X \cap \pi_{23}^{-1}\Delta_X)\big) \subseteq c(\pi_{13}(\delta)) = c(\Delta_X)$, where $\delta \subseteq X^3$ is the image of the triple diagonal, by (i). The last sentence is clear. □

Local isomorphisms. Let a_j be the sheafification for j and W_j the class of maps inverted by a_j. A map $f\colon X \to Y$ is *j-bidense* if $\mathrm{im}f$ is c-dense in Y and $\ker f \subseteq c(\Delta_X)$, i.e. the diagonal $X \to X \times_Y X$ is c-dense. A map is in W_j iff it is j-bidense [1, A4.3]; in particular a monomorphism is in W_j iff it is c-dense, and an epimorphism is in W_j iff its kernel pair lies in $c(\Delta_X)$. For $\mathcal{E} = \mathrm{Presh}(S)$ and a Grothendieck topology J this is the description of J-local isomorphisms as locally injective and locally surjective maps [6, III.7, Cor. 6]; see Lemma 5.1.

Words. Fix topologies j, j' with closure operators c, c'. A *word* is a composite $w = c_1 c_2 \cdots c_n$ with each $c_i \in \{c, c'\}$, c_n applied first; n is its length. We abbreviate $(cc')^k$ as $c^{(k)}$, a word of length $2k$ ending with c'; $c^{(0)}$ is the

identity. Since the closures are inflationary, $w(A) \subseteq w'(A)$ whenever w is obtained from w' by deleting letters; in particular $c^{(k)}(A) \subseteq c^{(k+1)}(A)$, and $c(A), c'(A) \subseteq c^{(1)}(A)$.

The join topology. Let $\mathcal{E}$ be a Grothendieck topos, so that every $\mathrm{Sub}(X)$ is a frame and unions of subobjects are stable under pullback. Let $j \vee j'$ be the join in the lattice of topologies and $\bar{c}$ its closure operator. Define the transfinite iterates $c^{(0)}(A) = A$, $c^{(\alpha+1)}(A) = c^{(1)}(c^{(\alpha)}(A))$ and $c^{(\lambda)}(A) = \bigcup_{\alpha<\lambda} c^{(\alpha)}(A)$ for limit λ; they increase with α, are constant from some ordinal γ_A on since $\mathrm{Sub}(X)$ is a set, commute with pullback, and $c^{(\alpha)}(c^{(\beta)}(A)) = c^{(\beta+\alpha)}(A)$.

Lemma 2.2. *$\bar{c}(A) = \bigcup_\alpha c^{(\alpha)}(A)$ for every subobject A.*

Proof. Write $c^\infty(A)$ for the union. Since $c, c' \leq \bar{c}$ and $\bar{c}$ is monotone and idempotent, $c^{(\alpha)}(A) \subseteq \bar{c}(A)$ for all α by induction, with unions at limits. Conversely c^∞ is a universal closure operator: it is inflationary and monotone; $c^\infty(A) = c^{(\gamma)}(A)$ for $\gamma = \gamma_A$ is a fixed point of $c^{(1)}$, hence of c and c' (as $c'(B) \subseteq c(c'(B)) = B$ for a fixed point B) and of c^∞; it commutes with pullback because each $c^{(\alpha)}$ does and pullback preserves unions; and it preserves binary intersections because each $c^{(\alpha)}$ does and, $\mathrm{Sub}(X)$ being a frame, $\bigcup_\alpha c^{(\alpha)}(A) \cap \bigcup_\beta c^{(\beta)}(B) = \bigcup_{\alpha,\beta} c^{(\alpha)}(A) \cap c^{(\beta)}(B) = \bigcup_\gamma c^{(\gamma)}(A \cap B)$. It lies above c and c', so its topology lies above j and j', hence above $j \vee j'$, and $\bar{c} \leq c^\infty$. □

Remark 2.3. The transfinite iterate is the "topological reflection" of [4, Cor. 4.2, Prop. 4.3], applied to the weak topology jj' instead of the pointwise join $j \vee_{\mathrm{w}} j'$; since $j \vee_{\mathrm{w}} j' \leq jj' \leq j \vee j'$, the two iterations have the same limit.

3. Uniform bidensity and the join

Fix topologies j, j' on a topos $\mathcal{E}$ with closure operators c, c', and write $V := W_j \vee W_{j'}$ and, when $\mathcal{E}$ is a Grothendieck topos, $K := j \vee j'$ with closure operator $\bar{c}$.

Definition 3.1. For $k \geq 0$ let U_k be the class of maps $f\colon X \to Y$ with $c^{(k)}(\mathrm{im} f) = Y$ and $\ker f \subseteq c^{(k)}(\Delta_X)$. Put $U = \bigcup_k U_k$.

U_0 is the class of isomorphisms (an epimorphism whose kernel pair is the diagonal is an isomorphism). A map in U_k is bidense for the join topology *uniformly*: the ordinal of Lemma 2.2 needed for its image and its kernel pair is the finite number k.

Lemma 3.2. *$U_k \subseteq U_{k+1}$, $W_j \cup W_{j'} \subseteq U_1$, U is a sameness, and in a Grothendieck topos $U \subseteq W_K$. Hence $V \subseteq U$, and $V \subseteq U \subseteq W_K$ in a Grothendieck topos.*

Proof. The first two assertions follow from $c^{(k)} \le c^{(k+1)}$ and $c, c' \le c^{(1)}$, the fourth from $c^{(k)} \le \bar{c}$ (Lemma 2.2). Identities lie in U_0. Let $h\colon X \to Y$ and $g\colon Y \to Z$; we use Lemma 2.1 throughout, for the words $c^{(k)}$.

Composition. Let $h \in U_k$, $g \in U_l$. Then $\mathrm{im}(gh) = g(\mathrm{im}h)$ and $c^{(k+l)}(\mathrm{im}gh) \supseteq c^{(l)}\big(g(c^{(k)}(\mathrm{im}h))\big) = c^{(l)}(g(Y)) = c^{(l)}(\mathrm{im}g) = Z$ by (i). And $\ker(gh) = (h\times h)^{-1}(\ker g) \subseteq (h\times h)^{-1}(c^{(l)}(\Delta_Y)) = c^{(l)}((h\times h)^{-1}\Delta_Y) = c^{(l)}(\ker h) \subseteq c^{(l+k)}(\Delta_X)$.

Cancellation of g. Let $gh \in U_k$ and $g \in U_k$. Then $\ker h \subseteq \ker(gh) \subseteq c^{(k)}(\Delta_X)$. Let $B := g^{-1}(\mathrm{im}gh) \subseteq Y$; then $c^{(k)}(B) = g^{-1}(c^{(k)}(\mathrm{im}gh)) = g^{-1}(Z) = Y$. Moreover $B = \pi_1\big(\ker g \cap (Y \times \mathrm{im}h)\big)$, and

$$\begin{aligned}\pi_1\big(\ker g \cap (Y\times \mathrm{im}h)\big) &\subseteq \pi_1\big(c^{(k)}(\Delta_Y) \cap (Y\times \mathrm{im}h)\big)\\ &\subseteq \pi_1\big(c^{(k)}(\Delta_Y \cap (Y\times \mathrm{im}h))\big)\\ &\subseteq c^{(k)}\big(\pi_1(\Delta_Y \cap (Y\times \mathrm{im}h))\big) = c^{(k)}(\mathrm{im}h)\end{aligned}$$

by (ii) and (i). Hence $Y = c^{(k)}(B) \subseteq c^{(2k)}(\mathrm{im}h)$.

Cancellation of h. Let $gh \in U_k$ and $h \in U_k$. Then $\mathrm{im}g \supseteq \mathrm{im}(gh)$, so $c^{(k)}(\mathrm{im}g) = Z$. Since $h\times h\colon X\times X \to \mathrm{im}h \times \mathrm{im}h$ is an epimorphism, $\ker g \cap (\mathrm{im}h \times \mathrm{im}h) = (h\times h)(\ker gh) \subseteq (h\times h)(c^{(k)}(\Delta_X)) \subseteq c^{(k)}(\Delta_Y)$ by (i); and $c^{(k)}(\mathrm{im}h \times \mathrm{im}h) = Y \times Y$ by (iii). So $\ker g = \ker g \cap c^{(k)}(\mathrm{im}h \times \mathrm{im}h) \subseteq c^{(k)}\big(\ker g \cap (\mathrm{im}h \times \mathrm{im}h)\big) \subseteq c^{(2k)}(\Delta_Y)$ by (ii). □

Lemma 3.3 (peeling)**.** *Let $w = c_1c_2\cdots c_n$ be a word with c_i the closure operator of $j_i \in \{j, j'\}$.*

(a) *If $f\colon X \to Y$ is an epimorphism with $\ker f \subseteq w(\Delta_X)$, then $f = f_1 \circ \cdots \circ f_n \circ u$ with u an isomorphism and f_i an epimorphism in W_{j_i}.*

(b) *If $f\colon X \to Y$ is a monomorphism with $w(X) = Y$, then $f = f_1 \circ \cdots \circ f_n \circ u$ with u an isomorphism and f_i a monomorphism in W_{j_i}.*

For $n \ge 1$ the isomorphism u can be absorbed into f_n.

Proof. Induction on n; for $n = 0$ the hypothesis says that f is an isomorphism. Write $w = w'c_n$. (a) Let $R := \ker f \cap c_n(\Delta_X)$, an equivalence relation on X by Lemma 2.1(iv), and let $q\colon X \twoheadrightarrow X/R$ be its quotient, so that $\ker q = R \subseteq c_n(\Delta_X)$ and $q \in W_{j_n}$. Since $R \subseteq \ker f$, $f = \bar{f} \circ q$ with $\bar{f}$ an epimorphism, and $\ker \bar{f} = (q\times q)(\ker f)$ because $q\times q$ is an epimorphism and $\ker f = (q\times q)^{-1}(\ker \bar{f})$. Now

$$\begin{aligned}(q\times q)(\ker f) &= (q\times q)\big(w'(c_n\Delta_X) \cap \ker f\big) \subseteq (q\times q)\big(w'(c_n\Delta_X \cap \ker f)\big)\\ &= (q\times q)(w'(R)) \subseteq w'\big((q\times q)(R)\big) = w'(\Delta_{X/R})\end{aligned}$$

by Lemma 2.1(ii) and (i), as $(q\times q)(\ker q) = \Delta_{X/R}$. Induction applied to $\bar{f}$ and w' gives the factorisation. (b) Regard X as a subobject of Y and let $Z := c_n(X) \subseteq Y$. The inclusion $X \hookrightarrow Z$ is c_n-dense (the closure of X in Z is $c_n(X)\cap Z = Z$ by pullback stability), hence in W_{j_n}, and $w'(Z) = w(X) = Y$. Induction applied to $Z \hookrightarrow Y$ and w' gives the factorisation. □

Proof of Theorem 1.1. Let C be the closure of $W_j \cup W_{j'}$ under composition. Then $C \subseteq V \subseteq U$ by Lemma 3.2. Let $f \in U_k$, $k \geq 1$, and factor $f = m \circ e$ with $e\colon X \twoheadrightarrow \mathrm{im}f$ and $m\colon \mathrm{im}f \hookrightarrow Y$. Then $\ker e = \ker f \subseteq c^{(k)}(\Delta_X)$ and $c^{(k)}(\mathrm{im}f) = Y$, and $c^{(k)}$ is a word of length $2k$ ending with c'. By Lemma 3.3, e is a composite of $2k$ epimorphisms and m a composite of $2k$ monomorphisms, alternately in $W_{j'}$ and W_j, beginning with $W_{j'}$. So $U \subseteq C$, and $C = V = U$. □

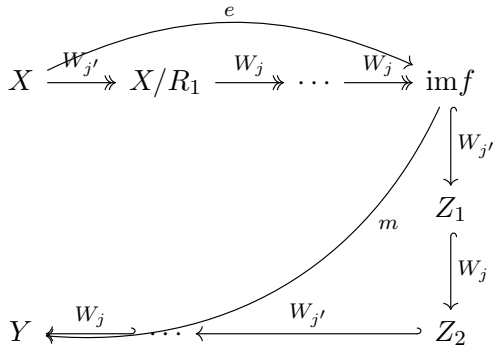


FIGURE 1. Theorem 1.1: a map in U_k is $2k$ quotients by "kernel pair intersected with the closure of the diagonal" followed by $2k$ alternating closures of its image, each step a local isomorphism for one of the two topologies.

Remark 3.4. In general the join of two samenesses requires the two-out-of-three saturation of the compositional closure [8]: on the poset $0 < 1 < 2$ viewed as a category, the classes $W_1 = \{0 \to 1\}$ and $W_2 = \{0 \to 2\}$ (with the identities) are samenesses, their compositional closure is $\{0 \to 1, 0 \to 2\}$, and two-out-of-three forces $1 \to 2$ into the join, which is therefore the class of all morphisms. For two classes of local isomorphisms no cancellation is needed, and this is the content of the peeling lemma rather than a formal consequence of the join theorem of [9, Thm. 4.1], which only identifies the join with the class inverted by a composite reflector when such a reflector exists. The number of steps is controlled in both directions: a composite of n maps of $W_j \cup W_{j'}$ lies in U_n by Lemma 3.2, and a map of U_k is a composite of $4k$ such maps. Theorem 1.1 and its proof hold in every elementary topos.

4. The criterion

Proposition 4.1. *Let $S \subseteq Y$ be a subobject, let $X_S := Y \amalg_S Y$ be the pushout of $Y \hookleftarrow S \hookrightarrow Y$ with injections ι_1, ι_2, and let $f_S\colon X_S \to Y$ be the fold map. For every word w (and for $\bar{c}$), $\ker f_S \subseteq w(\Delta_{X_S})$ iff $w(S) = Y$. Hence $f_S \in U_k$ iff $c^{(k)}(S) = Y$, and in a Grothendieck topos $f_S \in W_K$ iff S is $\bar{c}$-dense in Y.*

Proof. In a topos the pushout of a monomorphism along a monomorphism is a pullback square and its injections are monomorphisms [1, A2.4]; so ι_1, ι_2 are monomorphisms with intersection S, and the equaliser of ι_1, ι_2 is S. The map f_S is an epimorphism, as ι_1, ι_2 are jointly epimorphic, so its image is Y. Since $f_S\iota_1 = f_S\iota_2 = \mathrm{id}_Y$, the kernel pair of f_S is the union of the images of $(\iota_a, \iota_b)\colon Y \to X_S \times X_S$ over $a, b \in \{1, 2\}$, that is, of Δ_{X_S} and the images of (ι_1, ι_2) and (ι_2, ι_1). Now $(\iota_1, \iota_2)^{-1}(\Delta_{X_S})$ is the equaliser of ι_1, ι_2, i.e. S, so by pullback stability $(\iota_1, \iota_2)^{-1}(w(\Delta_{X_S})) = w(S)$, and the image of (ι_1, ι_2) lies in $w(\Delta_{X_S})$ iff $w(S) = Y$; the same holds for (ι_2, ι_1) by symmetry. The last two assertions follow with $w = c^{(k)}$ and $w = \bar{c}$. □

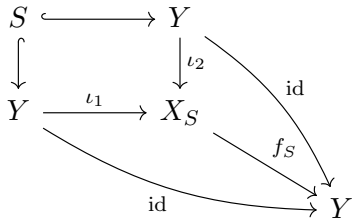


FIGURE 2. The witness of Proposition 4.1: two copies of Y glued along the subobject S; the kernel pair of the fold map is the diagonal together with the two copies of Y, which lie in $w(\Delta)$ exactly when $w(S) = Y$.

Proof of Theorem 1.2. The identity $(jj')^k = j \vee j'$ on Ω is equivalent to $c^{(k)} = \bar{c}$ on every $\mathrm{Sub}(X)$, since the closure of A for a topology t is classified by $t \circ \chi_A$ and the generic subobject $\top\colon 1 \to \Omega$ has $\chi_\top = \mathrm{id}_\Omega$. If $c^{(k)} = \bar{c}$, a map in W_K is K-bidense, hence in $U_k \subseteq V$, and $V \subseteq W_K$ always. If $c^{(k)} \neq \bar{c}$ for every k, choose for each k a subobject $S_k \subseteq Y_k$ with $c^{(k)}(S_k) \neq \bar{c}(S_k)$; replacing Y_k by $\bar{c}(S_k)$ we may assume S_k is $\bar{c}$-dense in Y_k but $c^{(k)}(S_k) \neq Y_k$, as $c^{(k)} \leq \bar{c}$ and both commute with pullback along $\bar{c}(S_k) \hookrightarrow Y_k$. Let $f_k := f_{S_k}\colon X_k \to Y_k$ and $F := \coprod_k f_k$. Then $F \in W_K$: it is an epimorphism, and since coproducts are disjoint and universal, $\ker F$ is the union of the images of $\ker f_k$ under the monomorphisms $X_k \times X_k \hookrightarrow (\coprod_k X_k)^2$, which lie in the images of $\bar{c}(\Delta_{X_k})$ by Proposition 4.1, hence in $\bar{c}(\Delta_{\coprod X_k})$ by Lemma 2.1(i). But $F \notin U_k$ for every k: pulling back along the summand $X_k \times X_k \hookrightarrow (\coprod X)^2$ and $Y_k \hookrightarrow \coprod Y$, the conditions for F restrict to those for f_k (pullback stability of $c^{(k)}$, disjointness of coproducts), and $f_k \notin U_k$ by Proposition 4.1. So $F \in W_K \setminus U = W_K \setminus V$ by Theorem 1.1. □

Remark 4.2. Lemma 2.2 and Theorem 1.2 place the question in the setting of [12, 14]: there the join $j_1 \vee j_2$ of two nuclei on a Heyting algebra or frame L is the transfinite iterate of $j_2 j_1$ [2], and the identity $(j_2 j_1)^N = j_1 \vee j_2$ for some N, the finite-iteration axiom of the lax logic of the join in [12], is exactly

the condition under which the join of the two reflections of L-fuzzy sets is generated at the level of inverted maps [14, Cor. 4.3]. Theorem 1.2 is the same statement for the internal frame Ω of a Grothendieck topos with the Lawvere–Tierney topologies j, j': the join of two subtopos classes is the class of the meet of the subtoposes iff the alternation of the two local operators reaches the join uniformly in finitely many steps. The condition is uniform in the subobject; it should be distinguished from the pointwise condition that the iteration terminates at each element, which suffices in $\mathrm{Same}(L)$ [14] and which, for the relativized necessities $\Box_C, \Box_{C'}$ on the frame of upsets of a poset, is the absence of infinite alternating chains characterising $\Box_C \vee \Box_{C'} = \Box_{C\cap C'}$ [13]; a disjoint union of finite alternating chains of unbounded length satisfies the pointwise condition but not the uniform one.

5. Presheaf toposes and topologies of ideal type

The sieve dictionary. Let $\mathcal{E} = \mathrm{Presh}(S)$ for a small category S. A sieve on c is a set R of morphisms with codomain c closed under precomposition; for $b\colon x \to c$, $b^*R = \{e : be \in R\}$. A Grothendieck topology J assigns to each c a set $J(c)$ of covering sieves [6, III.2], and corresponds to the Lawvere–Tierney topology j with $j(R) = \{b\colon x \to c :\ b^*R \in J(x)\}$ on $\Omega(c)$, the set of sieves on c; the closure of a subpresheaf $A \subseteq X$ is $c(A)(d) = \{x \in X(d) : \{b : x\cdot b \in A\} \in J\}$ [6, V.1, V.4]. A sieve R is covering iff $j(R)$ is the maximal sieve. For a map $f\colon X \to Y$, $y \in Y(d)$ and $x, x' \in X(d)$ put

$$\mathrm{Im}_f(y) = \{b\colon d_0 \to d :\ y\cdot b \in \mathrm{im} f_{d_0}\}, \quad \mathrm{Eq}(x,x') = \{b\colon d_0 \to d :\ x\cdot b = x'\cdot b\}.$$

Both are sieves, $b^*\mathrm{Im}_f(y) = \mathrm{Im}_f(y\cdot b)$ and $b^*\mathrm{Eq}(x,x') = \mathrm{Eq}(x\cdot b, x'\cdot b)$. For a word w in j, j' call a sieve R on d *w-reachable* if $\mathrm{id}_d \in w(R)$, i.e. $w(R)$ is maximal, and *k-reachable* if $(jj')^k$-reachable.

Lemma 5.1. *Let w be a word in c, c' and $\tilde{w}$ the corresponding word in j, j'. Then $y \in w(\mathrm{im} f)$ iff $\mathrm{Im}_f(y)$ is $\tilde{w}$-reachable, and $(x,x') \in w(\Delta_X)$ iff $\mathrm{Eq}(x,x')$ is $\tilde{w}$-reachable. Hence $f \in U_k$ iff all image sieves of f and all equaliser sieves $\mathrm{Eq}(x,x')$ with $f(x) = f(x')$ are k-reachable, and $f \in W_J$ iff these sieves are J-covering.*

Proof. By induction on the length of w, using that for a subpresheaf A of X and $x \in X(d)$ the sieve $\{b : x\cdot b \in A\}$ is, for $A = w_0(\mathrm{im} f)$, the sieve $\{b : \mathrm{Im}_f(y\cdot b)$ is $\tilde{w}_0$-reachable$\} = \{b : b^*\mathrm{Im}_f(y)$ is $\tilde{w}_0$-reachable$\} = \tilde{w}_0(\mathrm{Im}_f(y))$ by the induction hypothesis and pullback stability of $\tilde{w}_0$; so $y \in c_1(w_0(\mathrm{im} f))$ iff $\tilde{w}_0(\mathrm{Im}_f(y)) \in J_1$ iff $\mathrm{id} \in j_1\tilde{w}_0(\mathrm{Im}_f(y))$. The same computation applies to Eq with $A = w_0(\Delta_X)$ and the pairs $(x\cdot b, x'\cdot b)$. The last assertion is the bidense description, with $w = c$ and c', and is the classical one [6, III.7, Cor. 6]. □

Ideal topologies. For an ideal A the sets $A(-,c)$ are sieves and $A(-,c)\cdot b \subseteq A(-,x)$ for $b\colon x \to c$.

Lemma 5.2. *Let D, E be idempotent ideals.*

(i) *J_D is a topology, $D(-,c)$ is its least covering sieve on c, and $J_D \leq J_E$ iff $D \supseteq E$.*
(ii) *The closure operator of J_D is $j_D(S) = \{b : \ b \cdot D \subseteq S\}$, where $b \cdot D = \{b \circ d : d \in D(-, \operatorname{dom} b)\}$; hence $(j_D j_E)^k(S) = \{b : \ b \cdot (DE)^k \subseteq S\}$, and S is k-reachable for J_D, J_E iff $S \supseteq (DE)^k(-,c)$. A map is in U_k iff it is* locally bijective along $(DE)^k$*: $f(x) = f(x')$ implies $x \cdot a = x' \cdot a$ for all $a \in (DE)^k(-,c)$, and $y \cdot a \in \operatorname{im} f$ for all $y \in Y(c)$ and $a \in (DE)^k(-,c)$.*
(iii) *For every k and c the sieve $(DE)^k(-,c)$ is $(J_D \vee J_E)$-covering.*
(iv) *If $(DE)^m = (DE)^{m+1} =: I'$ then I' is idempotent, $I' \subseteq D \cap E$, $DI' = I'D = EI' = I'E = I'$, $(ED)^{m+1} = I'$, every alternating product of at least $2(m+1)$ factors D, E, in either order, equals I', and $J_D \vee J_E = J_{I'}$. (The chain $(ED)^k$ need not have stabilised at index m: Example 5.4.)*

Proof. (i) is [9, Thm. 5.1]: the maximal sieve contains $D(-,c)$; pullback of a sieve containing $D(-,c)$ along $b\colon x \to c$ contains $D(-,x)$ because D is a left ideal; and if $R \supseteq D(-,c)$ and each a^*R' with $a \in R$ contains $D(-, \operatorname{dom} a)$, then for $d \in D(-,c)$, $d = d_1 d_2$ with $d_i \in D$ by idempotency, $d_1 \in R$ and $d_2 \in d_1^* R'$, so $d \in R'$. (ii) $b \in j_D(S)$ iff $b^*S \supseteq D(-, \operatorname{dom} b)$ iff $b \cdot D \subseteq S$; then $b \in j_D j_E(S)$ iff $(b \circ d) \cdot E \subseteq S$ for all d iff $b \cdot DE \subseteq S$, and inductively for $(DE)^k$; $\mathrm{id}_c \cdot (DE)^k = (DE)^k(-,c)$. For a map f, $\mathrm{Im}_f(y) \supseteq (DE)^k(-,c)$ says $y \cdot a \in \operatorname{im} f$ for $a \in (DE)^k(-,c)$, and $\mathrm{Eq}(x,x') \supseteq (DE)^k(-,c)$ says $x \cdot a = x' \cdot a$; now apply Lemma 5.1. (iii) follows from Lemma 2.2: $(DE)^k(-,c)$ is k-reachable, and a k-reachable sieve is K-covering since $(jj')^k \leq j_K$. (iv) $I'I' = (DE)^{2m} = I'$; $I' \subseteq DE \subseteq D \cap E$; $DI' = DD\,E(DE)^{m-1} = (DE)^m = I'$ as $DD = D$, and similarly $I'E = I'$; $I'D \supseteq I'I' = I'$ as $I' \subseteq D$, and $I'D \subseteq I'$ as I' is a right ideal, so $I'D = I'$, and likewise $EI' = I'$; then $(ED)^{m+1} = E(DE)^m D = EI'D = I'$. An alternating product of at least $2(m+1)$ factors contains $(DE)^{m+1}$ or $(ED)^{m+1}$ as a block of consecutive factors; the block equals I', and multiplying I' by D or E on either side returns I'. For the join, $J_{I'} \geq J_D, J_E$ since $I' \subseteq D, E$, and any topology above J_D and J_E has $I'(-,c) = (DE)^m(-,c)$ covering by (iii), hence is above $J_{I'}$. □

Proof of Corollary 1.3. By Lemma 5.2(ii) and Theorem 1.1, $W_{J_D} \vee W_{J_E}$ is the class of maps locally bijective along some $(DE)^k$. If $(DE)^m = (DE)^{m+1} = I'$, then $K = J_{I'}$ by Lemma 5.2(iv), a K-covering sieve contains $I'(-,c) = (DE)^m(-,c)$ and is therefore m-reachable; Theorem 1.2 gives $V = W_K = W_{J_{I'}}$, and Theorem 1.1 gives the factorisation into $4m$ maps, since $W_{J_{I'}} = U_m$. If the chain does not stabilise, then $(DE)^k \neq (DE)^{k+1}$ for every k (once two consecutive terms agree, all later ones do), so there are c_k with $(DE)^k(-,c_k) \not\subseteq (DE)^{k+1}(-,c_k)$; the sieve $S_k = (DE)^{k+1}(-,c_k)$ is K-covering by Lemma 5.2(iii) and not k-reachable by Lemma 5.2(ii), so Theorem 1.2 gives $V \neq W_K$. The witnesses of Proposition 4.1 are here two

copies of $y(c_k)$ glued along $(DE)^{k+1}(-, c_k)$, whose fold map is locally bijective along $(DE)^{k+1}$ but identifies the two identities, which restrict along a morphism of $(DE)^k \setminus (DE)^{k+1}$ into c_k to distinct elements. □

Remark 5.3. The first version of this paper proved the stabilising direction by the route of [11]: $2r$ alternating separated quotients, $r = m + 1$, followed by $4r$ alternating plus constructions, with a retraction of the representing bimodule $I' \otimes I'$ onto the alternating tensor power. Theorem 1.1 replaces this by the single observation $W_{J_{I'}} = U_m \subseteq V$ and improves the bound to $4m$ maps. The plus tower is nevertheless of independent interest, as it need not stabilise even when the chain does [11, §7].

Example 5.4. The two chains $(DE)^k$ and $(ED)^k$ need not stabilise at the same index. Let S be the category with two objects $0, 1$ and one non-identity arrow $a\colon 0 \to 1$, and let $D = \{\mathrm{id}_0, a\}$, $E = \{\mathrm{id}_1, a\}$ be the ideals of arrows factoring through 0, respectively through 1; both are idempotent. Then $DE = \emptyset$ (no arrow of E has codomain 0), while $ED = \{a\}$. So $(DE)^1 = (DE)^2 = \emptyset = I'$, $m = 1$, but $(ED)^1 = \{a\} \neq I'$, and only $(ED)^2 = \emptyset = I'$. The join $J_D \vee J_E = J_\emptyset$ is the degenerate topology, as the empty sieve on 0 is J_E-covering and the sieve $\{a\}$ on 1 is J_D-covering.

Example 5.5. On a finite category every chain of ideals stabilises, so Corollary 1.3 contains [11, Thm. 1.1]. On $S = (\mathbb{N}, \leq)$ with D and E the ideals of arrows $n \to m$ passing through an even, respectively an odd, point, $(DE)^k$ consists of the arrows $n \to m$ admitting k alternating even and odd points between n and m, a strictly decreasing chain with empty intersection; the join topology is degenerate and W_K is everything, while V is not [10, Thm. 7.1]. Example 6.3 describes V.

Example 5.6. For a monoid M, viewed as a category with one object, ideals are the two-sided ideals of M and the criterion is the stabilisation of the alternating products $DE \supseteq DEDE \supseteq \cdots$ in the quantale of ideals of M. Every finite monoid satisfies it, with m at most the number of elements. For infinite monoids the chain can decrease forever, and then the join of the two classes of local isomorphisms of M-sets is the class of maps locally bijective along some $(DE)^k$, strictly smaller than the class for the join topology.

6. Ideals generated by objects

For a set A of objects of S let D_A be the set of morphisms factoring through an object of A. It is an ideal, idempotent since $f = g \circ h$ with $h\colon x \to a$, $g\colon a \to y$ has $g, h \in D_A$; on a finite category every idempotent ideal is of this form after Cauchy completion [11, §2], but not in general: on $(\mathbb{Q}, \leq)$ the ideal of strict inequalities is idempotent and contains no identity. Write $W_A := W_{J_{D_A}}$.

Proposition 6.1. *Let A, B be sets of objects of S.*

(a) *W_A is the class of maps f with f_a bijective for every $a \in A$; $(D_A D_B)^k$ is the set of morphisms factoring through a chain $b_1 \to a_1 \to b_2 \to \cdots \to b_k \to a_k$ with $b_i \in B$, $a_i \in A$, and contains $D_{A\cap B}$. The join $W_A \vee W_B$ is the class of maps f such that, for some k, f is bijective along every such chain: if $f(x) = f(x')$ then $x \cdot g = x' \cdot g$, and every $y \cdot g$ lies in the image, for every morphism g factoring through a chain of length k.*

(b) *$W_A \vee W_B = W_{A\cap B}$ if and only if for some N every morphism factoring through such a chain of length N factors through an object of $A \cap B$, i.e. $(D_A D_B)^N \subseteq D_{A\cap B}$.*

(c) *If $S = P$ is a poset whose intervals are finite, the condition in (b) holds if and only if the strictly alternating chains $b_1 < a_1 < \cdots < b_k < a_k$ ($b_i \in B$, $a_i \in A$) contained in intervals disjoint from $A \cap B$ have a bounded number of elements; if they do not, the chain $(D_A D_B)^k$ does not stabilise and $W_A \vee W_B \subsetneq W_{J_{D_A} \vee J_{D_B}}$.*

Proof. (a) If f_a is bijective for $a \in A$, then for $x, x' \in X(c)$ with $f(x) = f(x')$ and $g = g_1 g_2 \in D_A(-, c)$ with $g_1 \colon a \to c$, $a \in A$, we get $f_a(x \cdot g_1) = f_a(x' \cdot g_1)$, so $x \cdot g_1 = x' \cdot g_1$ and $x \cdot g = x' \cdot g$; and for $y \in Y(c)$, $y \cdot g_1 = f_a(z)$ for some z, so $y \cdot g = f(z \cdot g_2)$. Thus f is locally bijective along D_A, and $f \in W_A$ by Lemma 5.2(ii). Conversely $\mathrm{id}_a \in D_A(-, a)$ for $a \in A$, so local bijectivity along D_A gives f_a bijective. A morphism of $D_A D_B$ is $d \circ e$ with e factoring through some $b \in B$ and d through some $a \in A$, i.e. a morphism factoring through a chain $b \to a$; iterating gives the description of $(D_A D_B)^k$, and a morphism through $c \in A \cap B$ factors through the chain $c \to c \to \cdots \to c$ of identities. The description of the join is Corollary 1.3. (b) If $(D_A D_B)^N \subseteq D_{A\cap B}$ then $(D_A D_B)^N \subseteq D_{A\cap B} \subseteq (D_A D_B)^{N+1} \subseteq (D_A D_B)^N$, so the chain stabilises at $I' = D_{A\cap B}$ and Corollary 1.3 gives $W_A \vee W_B = W_{J_{I'}} = W_{A\cap B}$. Conversely, $D_{A\cap B} \subseteq D_A \cap D_B$ gives $J_{D_{A\cap B}} \geq K := J_{D_A} \vee J_{D_B}$ by Lemma 5.2(i), hence $W_{A\cap B} \supseteq W_K \supseteq W_A \vee W_B$. If $W_A \vee W_B = W_{A\cap B}$, then $W_A \vee W_B = W_K$, the chain stabilises at some $I' = (D_A D_B)^m$ by Corollary 1.3, and $W_{J_{I'}} = W_K = W_{J_{D_{A\cap B}}}$. A topology is determined by the class of maps inverted by its sheafification, since the sheaves are the objects local with respect to that class (a reflective subcategory is recovered from the maps its reflector inverts) and subtoposes of $\mathrm{Presh}(S)$ correspond bijectively to topologies [6, Chap. V]; so $J_{I'} = J_{D_{A\cap B}}$, $I' = D_{A\cap B}$ by Lemma 5.2(i), and $N = m$ works. (c) In a poset, $(n \leq m) \in (D_A D_B)^k$ iff there are $n \leq b_1 \leq a_1 \leq \cdots \leq b_k \leq a_k \leq m$ with $b_i \in B$, $a_i \in A$; if the interval $[n, m]$ is disjoint from $A \cap B$, an equality $b_i = a_i$ or $a_i = b_{i+1}$ would produce a point of $A \cap B$, so the chain is strict. Hence $(D_A D_B)^k \setminus D_{A\cap B}$ consists of the pairs $n \leq m$ with $[n, m]$ disjoint from $A \cap B$ and containing a strictly alternating chain with $2k$ elements. If these chains have at most L elements, then $(D_A D_B)^k = D_{A\cap B}$ for $2k > L$. If their sizes are unbounded, $(D_A D_B)^k \neq D_{A\cap B}$ for every k, while $\bigcap_k (D_A D_B)^k = D_{A\cap B}$ because a chain with more elements than the finite interval $[n, m]$ is not strict; a descending chain whose members all

differ from its intersection does not stabilise, and Corollary 1.3 gives the strict inclusion. □

Example 6.2. On $P = (\mathbb{N}, \leq)$ let A be the even numbers and B the odd numbers together with the multiples of 4, so $A \cap B = 4\mathbb{N}$. An interval disjoint from $4\mathbb{N}$ lies in some $[4j+1, 4j+3]$, whose longest strictly alternating chain is $4j+1 < 4j+2 < 4j+3$; so $L = 3$, $(D_A D_B)^2 = D_{4\mathbb{N}}$, and a map of presheaves on $\mathbb{N}$ lies in the join of "bijective at the even points" and "bijective at the odd points and at the multiples of 4" if and only if it is bijective at the multiples of 4.

Example 6.3. On $P = (\mathbb{N}, \leq)$ let A be the even and B the odd numbers. A presheaf is a tower $\cdots \to X(2) \to X(1) \to X(0)$ of sets, and $(D_A D_B)^k(-, c)$ consists of the $n \leq c$ with $c - n \geq 2k - 1$, or $\geq 2k$ when n is even. By Proposition 6.1(a), $W_A \vee W_B$ is the class of maps of towers $f\colon X \to Y$ that are *bijective up to a shift*: for some s, $f_c(x) = f_c(x')$ implies $x|_{c-s} = x'|_{c-s}$, and every $y \in Y(c)$ has $y|_{c-s} \in \mathrm{im} f_{c-s}$ (for $c \geq s$; for $c < s$ the conditions are void). Such a map becomes an isomorphism of pro-objects: it has the inverse up to shift $g_n\colon Y(n+2s) \to X(n)$, $y \mapsto x|_n$ for any $x \in X(n+s)$ with $f(x) = y|_{n+s}$, which is well defined and compatible with restriction. The converse fails: a pro-isomorphism of towers only needs, for each n, *some* later stage $m(n)$ at which the inverse is defined, and the join requires the uniform bound $m(n) = n + 2s$. For instance let $X(n) = \{0, 1\}$ for all n, with the restriction $X(n+1) \to X(n)$ constant with value 0 when n is a power of 2 and the identity otherwise, and let $f\colon X \to 1$ be the map to the terminal tower. The section $1 \to X$ picking 0 is a pro-inverse, since for every n the composite restriction from some later stage to n is constant; but on an interval $[n, n+s]$ containing no power of 2 the two elements of $X(n+s)$ are identified by f and remain distinct at n, so f is bijective up to no uniform shift and $f \notin W_A \vee W_B$ (although $f \in W_K$, as W_K is the class of all maps). So in Same(Presh($\mathbb{N}$)) the join of "bijective at the even points" and "bijective at the odd points" is a uniform version of pro-isomorphism, strictly between the two classes and the pro-isomorphisms. The obstruction $\lim_{\mathbb{N}^{\mathrm{op}}}$ used in [10, Thm. 7.1] inverts every pro-isomorphism, and more (the projection $\{m \geq n\} \times \{0, 1\} \to \{m \geq n\}$ has empty limits on both sides and is no pro-isomorphism), so it separates the join from W_K without describing it: $W_A \vee W_B \subsetneq W_{\text{pro-iso}} \subsetneq W_{\lim} \subseteq W_K$.

Remark 6.4. For a set A of objects, the restriction functor along the full inclusion $i\colon A \hookrightarrow S$ preserves limits and has a fully faithful right adjoint, the right Kan extension Ran_i (and a left adjoint, the left Kan extension). Its essential image is a reflective subcategory whose reflector inverts exactly W_A, hence it is the subcategory of J_{D_A}-sheaves: $\mathrm{Sh}(S, J_{D_A}) \simeq \mathrm{Presh}(A)$, an essential subtopos of $\mathrm{Presh}(S)$ in the sense of [5]. The meet of the subtoposes $\mathrm{Presh}(A)$ and $\mathrm{Presh}(B)$ is the subtopos of $(J_{D_A} \vee J_{D_B})$-sheaves. Proposition 6.1(b) says that $W_A \vee W_B = W_{A \cap B}$ iff this meet is equivalent to $\mathrm{Presh}(A \cap B)$ *and* is reached by a bounded alternation; the first condition

alone does not suffice: for the even and the odd numbers on $\mathbb{N}$ the meet is equivalent to the degenerate subtopos $\mathrm{Presh}(\emptyset)$, yet the join of the samenesses is not $W_\emptyset$ but the class of maps bijective up to a uniform shift.

7. Localic toposes

Let L be a frame and $\mathrm{Sh}(L)$ its topos of sheaves. The Lawvere–Tierney topologies on $\mathrm{Sh}(L)$ correspond, as an ordered set, to the nuclei on L (the sublocales of L) [3, II.2], [1, A4.5]: a nucleus is a map $n\colon L \to L$ that is inflationary, monotone, idempotent and preserves finite meets, and the join $n \vee n'$ in the frame of nuclei corresponds to the join of the topologies. For $U \in L$ the subobjects of the representable sheaf $y(U)$ are the $V \leq U$, so $\Omega(U) = {\downarrow}U$.

Lemma 7.1. *Let j be the topology of the nucleus n and c its closure operator. For $V \leq U$ in L, the closure of V in $y(U)$ is $n(V) \wedge U$. Hence for nuclei n, n' and $V \leq U$, $(cc')^k(V) = (nn')^k(V) \wedge U$ in $\mathrm{Sub}(y(U))$, and the closure for $j \vee j'$ is $(n \vee n')(V) \wedge U$.*

Proof. The sheafification for j is the direct image along the sublocale inclusion followed by its inverse image, so a subobject $V \hookrightarrow W$ of $y(U)$ is j-dense iff $n(V) = n(W)$; the largest $W \leq U$ with $n(W) = n(V)$ is $n(V) \wedge U$, since $n(n(V) \wedge U) = n(V) \wedge n(U) = n(V)$ and any such W satisfies $W \leq n(W) = n(V)$. Thus $c(V) = n(V) \wedge U$. If $W = (nn')^k(V) \wedge U$ then $c'(W) = n'(W) \wedge U = n'((nn')^k V) \wedge n'(U) \wedge U = n'((nn')^k V) \wedge U$ and $c(c'(W)) = n(n'((nn')^k V)) \wedge n(U) \wedge U = (nn')^{k+1}(V) \wedge U$, as $U \leq n(U), n'(U)$. The last assertion is the first one for $n \vee n'$. □

Corollary 7.2. *For nuclei n, n' on a frame L, $W_n \vee W_{n'} = W_{n \vee n'}$ in $\mathrm{Same}(\mathrm{Sh}(L))$ if and only if $(nn')^k = n \vee n'$ for some k.*

Proof. By Theorem 1.2 and Lemma 7.1, the join is preserved iff there is k with $(nn')^k(V) \wedge U = (n \vee n')(V) \wedge U$ for all $V \leq U$; taking $U = \top$ gives $(nn')^k = n \vee n'$, and conversely. □

Example 7.3. For $U, U' \in L$ let $o(U)(V) = U \to V$ and $c(U')(V) = U' \vee V$ be the open and the closed nucleus. The composite $o(U)c(U')$, $V \mapsto U \to (U' \vee V)$, is idempotent: $U \to (U' \vee (U \to W)) = U \to W$ for $W \geq U'$, because $U \wedge (U' \vee (U \to W)) \leq W$. It is therefore a nucleus above both, and the least one, so $o(U) \vee c(U') = o(U)c(U')$ with $k = 1$; likewise $o(U)o(U') = o(U \wedge U')$ and $c(U)c(U') = c(U \vee U')$. For open and closed sublocales the join of the two classes is always the class of the meet of the sublocales.

Example 7.4. Let $X = [0, 1]$ with the open sets $\emptyset$, X and $(a, 1]$ for $0 \leq a < 1$; its frame of opens is the chain $[0, 1]$ (via $(a, 1] \mapsto 1 - a$, $\emptyset \mapsto 0$), a spatial frame in which no element other than the top is a join of completely join-prime elements, so $\mathrm{Sh}(X)$ is not a presheaf topos. A nucleus on a chain is determined by its set F of fixed points, closed under meets and

containing the top, $n_F(x) = \min\{f \in F : f \geq x\}$, and $n_F \vee n_G = n_{F\cap G}$. Take $F = \{0,1\} \cup \{1/2k : k \geq 1\}$ and $G = \{0,1\} \cup \{1/(2k+1) : k \geq 1\}$ [14, Ex. 5.3]. Then $n_F \vee n_G = n_{\{0,1\}}$, the double-negation nucleus of the chain (its fixed points form the two-element frame, so its sublocale is a single point and the corresponding subtopos is **Set**), but for $x = 1/2m$ one has $n_F n_G(x) = 1/(2m-2)$, so $(n_F n_G)^k(x) < 1$ for $k < m-1$: the identity $(n_F n_G)^k = n_F \vee n_G$ fails for every k, although $(n_F n_G)^k(x)$ reaches $(n_F \vee n_G)(x)$ for each fixed x. By Corollary 7.2, in $\mathrm{Same}(\mathrm{Sh}(X))$ the join of the two classes of local isomorphisms is strictly smaller than the class of maps inverted by the double-negation sheafification; by Theorem 1.1 it is the class of maps whose image and kernel pair are reached in a bounded number of alternating closures. With $F = \{1\} \cup \{1 - 1/2k\}$ and $G = \{1\} \cup \{1 - 1/(2k+1)\}$ the identity fails even pointwise. Finite fixed sets give $k \leq |F| + |G|$, so the join is preserved.

Subspaces. Let X be a topological space and $A \subseteq X$ a subspace with inclusion $i\colon A \to X$; we write i also for the induced geometric morphism $\mathrm{Sh}(A) \to \mathrm{Sh}(X)$, with inverse image i^*. The image of this geometric morphism is the subtopos of the sublocale determined by A, whose nucleus is $n_A(V) = \bigcup\{W \in O(X) : W \cap A \subseteq V\} = \mathrm{int}(V \cup (X \setminus A))$ [3, II.2]; since the inverse image of a geometric surjection is conservative, $W_A := W_{n_A}$ is the class of maps f of sheaves on X with i^*f an isomorphism. On complements, writing $V = X \setminus F$ with F closed, $n_A(X \setminus F) = X \setminus \mathrm{cl}(F \cap A)$.

Proposition 7.5. *Let $A, B \subseteq X$ be subspaces and, for closed $F \subseteq X$, let $T(F) = \mathrm{cl}(\mathrm{cl}(F \cap B) \cap A)$, with transfinite iterates T^α (intersections at limits). Then $(n_A n_B)^k(X \setminus F) = X \setminus T^k(F)$, the join nucleus is $X \setminus F \mapsto X \setminus T^\infty(F)$, and $W_A \vee W_B = W_{n_A \vee n_B}$ in $\mathrm{Same}(\mathrm{Sh}(X))$ iff there is k with $T^k(F) = T^\infty(F)$ for every closed F. If A and B are both closed then $T(F) = F \cap A \cap B$, if both are open then $n_A n_B = n_{A\cap B}$, and in both cases $k = 1$.*

Proof. The formula for $(n_A n_B)^k$ is the formula for n_A, n_B on complements; the join nucleus is the transfinite iterate by Lemma 2.2 (unions of opens are intersections of closed sets), and the criterion is Corollary 7.2. For closed A, B, $F \cap B$ and $\mathrm{cl}(F \cap B) \cap A$ are closed; for open A, B, n_A is the open nucleus $V \mapsto A \to V$, and $o(A)o(B) = o(A \cap B)$. □

Example 7.6. Let $X = \mathbb{R}$, $A = \mathbb{Q}$ and $B = \mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$. Both are dense, so both sublocales contain the least dense sublocale of $\mathbb{R}$ [3, II.2], and the meet $\mathbb{Q} \wedge \mathbb{I}$ of the two sublocales is a nonempty dense sublocale although the subspaces are disjoint. Here $T(F) = \mathrm{cl}(\mathrm{cl}(F \cap \mathbb{I}) \cap \mathbb{Q})$. Let C be a countable closed subset of $\mathbb{R}$ of finite Cantor–Bendixson rank r whose points of even rank are rational and whose points of odd rank are irrational. Such sets exist for every r: start with $C_0 = \{0\}$ and, given C_r, choose $t = 0$ if $r+1$ is even and $t = \sqrt{2}$ if $r+1$ is odd, disjoint intervals I_n of length tending to 0 accumulating only at t, and rationals p_n, $q_n > 0$ with $p_n + q_n C_r \subseteq I_n$; then

$C_{r+1} = \{t\} \cup \bigcup_n (p_n + q_n C_r)$ is countable and closed, the affine maps with rational coefficients preserve rationality, and t is the unique point of rank $r+1$. Since C is scattered, the isolated points of each derived set $C^{(m)}$, which are exactly the points of rank m, are dense in $C^{(m)}$; hence every point of rank $m+1$ is a limit of points of rank m. Therefore $\mathrm{cl}(C \cap \mathbb{I}) = C'$: the isolated points of C are rational, every point of odd rank is irrational, and every point of even rank ≥ 2 is a limit of irrational points of rank one less; likewise $\mathrm{cl}(C' \cap \mathbb{Q}) = C''$, and in general $\mathrm{cl}(C^{(m)} \cap \mathbb{I}) = C^{(m+1)}$ for even m and $\mathrm{cl}(C^{(m)} \cap \mathbb{Q}) = C^{(m+1)}$ for odd m; so $T^j(C) = C^{(2j)}$, the $2j$-th derived set, which is nonempty for $2j < r$, while $T^\infty(C) = \emptyset$ because iterated derived sets of a countable closed set are eventually empty. Taking $r > 2k$ shows that $(n_{\mathbb{Q}} n_{\mathbb{I}})^k \neq n_{\mathbb{Q}} \vee n_{\mathbb{I}}$ for every k: by Proposition 7.5, in $\mathrm{Same}(\mathrm{Sh}(\mathbb{R}))$ the join of the class of maps of sheaves that are isomorphisms over $\mathbb{Q}$ and the class of those that are isomorphisms over the irrationals is strictly smaller than the class of maps inverted by the sheafification onto the subtopos $\mathrm{Sh}(\mathbb{Q}) \wedge \mathrm{Sh}(\mathbb{I})$; by Theorem 1.1 it is the class of maps whose image and kernel pair are reached by a bounded number of the alternating derivatives T.

8. Remarks and questions

Remark 8.1. Theorem 1.1 holds in every elementary topos. Theorem 1.2 uses, for the identification of the closure operator of $j \vee j'$ with the transfinite iterate, that subobject lattices are frames and unions are pullback-stable, and for the necessity of a uniform k a countable coproduct of witnesses; in an elementary topos one still has, for each fixed k, that $W_{j \vee j'} \subseteq U_k$ iff $c^{(k)}$ is the closure operator of $j \vee j'$, by the same proof.

Remark 8.2. The classical sufficient condition for $W_j \vee W_{j'} = W_{j \vee j'}$ is that the tower of alternating sheafifications $(a_{j'} a_j)^k X$ reach $\mathrm{Sh}_{j \vee j'}(\mathcal{E})$ for one k and all X [9, Thm. 4.1], since then $(a_{j'} a_j)^k$ is the reflector onto the meet of the two subtoposes and its unit is a composite of maps of $W_j \cup W_{j'}$. Theorem 1.2 shows that the relevant iteration is that of the closure operators, not of the reflectors, and the two differ: on the contracted monoid of order 8 of [11, §7], with the ideals D, E generated by its two non-trivial idempotents, the chain $(DE)^k$ stabilises at $m = 2$ with $I' = \{0\}$, so the join of the two classes is preserved by Corollary 1.3, whereas the strictly alternating plus tower never reaches the $J_{I'}$-sheaves uniformly [11, Prop. 7.1], and a computation on all right M-sets with at most three elements (`scripts/comparison/sheaf_tower.py` in the repository) shows that neither $(a_E a_D)^k X$ nor $(a_D a_E)^k X$, $k \leq 2$, nor any other composite of at most eight plus constructions, is a $J_{I'}$-sheaf for all of them.

Question 8.3. Theorem 1.2 reduces join preservation to the identity $(jj')^k = j \vee j'$ on Ω. Which pairs of topologies on a given infinite category satisfy it? For ideal topologies this is the chain condition of Corollary 1.3

and for object-generated ideals the zigzag condition of Proposition 6.1; a structural description for topologies that are not of ideal type, for instance in terms of the two subtoposes, is open.

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